ModelIt!

Teacher's Guide

Why Can't Two Plants Share the Same Sunlight?

Modeling Competition for Limited Resources in Ecosystems

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-LS2-1, MS-LS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS2-1, MS-LS2-2

Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem. Construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how two plant species compete for limited sunlight and soil nutrients, with resource depletion creating feedback that determines which species thrives.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems** — Organisms and populations of organisms are dependent on their environmental interactions both with other living things and with nonliving factors. Competition for resources influences organism growth, survival, and reproduction.
Crosscutting Concept	**Cause and Effect** — Students investigate how asymmetric resource capture causes one species to dominate another, identifying the feedback mechanisms that amplify small initial advantages into large competitive outcomes.

CAST Testing Alignment

MULTIPLE CHOICE:

In a field, Species A grows 2 cm taller per week than Species B. After 8 weeks, Species A is significantly taller and Species B is wilting. Based on the competition model, what is the most likely cause of Species B's decline?

CONSTRUCTED RESPONSE:

Using your competition model, explain why invasive plant species often outcompete native plants when introduced to a new ecosystem. Reference sunlight capture, soil nutrient depletion, and the feedback loop between growth and resource availability.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Competition is one of the fundamental forces shaping ecosystems. When two organisms need the same limited resource, the outcome of their competition determines which species thrive, which decline, and how communities are structured. Understanding competition reveals why ecosystems look the way they do.

Competition for Light Is Asymmetric

Unlike nutrients or water, light comes from above. This means competition for light is inherently asymmetric: a taller plant shades a shorter one, but the shorter plant cannot shade the taller one. This asymmetry means that any height advantage compounds over time. The taller plant gets more light, grows more, gets even taller, and shades the shorter plant even more. This positive feedback loop is why forests have distinct canopy layers.

The Competitive Exclusion Principle

Ecologist G.F. Gause demonstrated in 1934 that two species competing for the exact same resource in the exact same way cannot coexist indefinitely. One will always eventually outcompete the other. This is why no two species occupy exactly the same ecological niche. When species do coexist, they have evolved to partition resources: different root depths, different light requirements, different growing seasons.

Resource Depletion Feedback

As competing organisms consume resources, those resources become scarcer. Scarcity intensifies competition because organisms must work harder (grow taller, extend roots deeper) to access what remains. This creates a feedback loop: growth depletes resources, scarcity intensifies competition, competition favors the stronger competitor, and the weaker competitor declines further. This feedback is why competition often leads to clear winners and losers.

Coexistence Through Niche Partitioning

Despite the competitive exclusion principle, ecosystems are full of biodiversity. Species coexist by partitioning resources: different depths in the soil, different heights in the canopy, different times of day for activity, or different types of nutrients. Some plants are shade-tolerant, thriving in the understory with less light. Others are sun-loving and dominate the canopy. This diversity of strategies allows many species to share an ecosystem.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify sunlight available and soil nutrients as the two external components, and Plant A growth, Plant B growth, and resource depletion as the three internal responses of the competition system.

E **Establish Relationships**

Students determine that sunlight positively drives both plant growth rates, soil nutrients positively drive both plant growth rates, both growth rates positively increase resource depletion, and resource depletion negatively reduces available sunlight and nutrients in a feedback loop.

V **Visualize & Model**

Students build a model showing how two plants draw from shared resource pools, with growth creating resource depletion that feeds back to limit both plants, especially the weaker competitor.

E **Evaluate Outcomes**

Students run equal-start and one-plant-advantage scenarios to observe how small initial differences amplify through the competition feedback loop.

R **Revise & Extend**

Students add water availability or root depth difference to explore how resource partitioning changes competition outcomes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"There is enough sunlight hitting this forest to power a small city. So why are the plants on the forest floor so small and struggling?"*

Watch For

Show a photo of a dense forest canopy with dim understory. Let students observe the light difference.

Slide 2: Learning Objectives

3 min

Say: *"Today we are modeling what happens when two plants fight for the same sunlight and nutrients. Spoiler: it is not a fair fight."*

Watch For

Have students read objectives. Pre-teach 'interspecific competition' and 'competitive exclusion.'

Slide 3: Big Question

3 min

Say: *"If I put two plants side by side with the same soil and light, will they both grow equally? What if one starts just a little taller?"*

Watch For

Think-pair-share: What do you think will happen to two identical plants growing next to each other?

Slide 4: LEVER Framework

2 min

Say: *"Our model has a hidden feedback loop. When one plant grows, it takes resources from the other, which makes the first plant grow even more."*

Watch For

Sketch the feedback loop: growth captures resources, which starves the competitor, which frees more resources for the winner.

Slide 5: Activity 1: Components

8 min

Say: *"Sunlight and soil nutrients are given by the environment. But plant growth and resource depletion are driven by the competition itself."*

Watch For

Guide sorting. Discuss why both plants draw from the same resource pools.

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the critical connection: Plant A's growth DEPLETES resources, which REDUCES Plant B's growth. And Plant B's growth does the same to Plant A."*

Watch For

Trace the competition loop. Both plants feed resource depletion, which hurts both, but hurts the weaker one more.

Slide 7: Activity 3: Simulation

10 min

Say: *"What if Plant A starts just 1 centimeter taller? Does that tiny advantage matter? Let's find out."*

Watch For

Students predict outcomes. Run equal start and slight advantage scenarios. Watch how small differences amplify.

Slide 8: Discoveries

5 min

Say: *"Even a tiny advantage compounds over time because the winner captures more resources, grows more, and captures even more. It is a snowball effect."*

Watch For

Connect to real examples: invasive species, forest succession, why weeding a garden helps your plants.

Slide 9: STEM Challenge

12 min

Say: *"Can you outsmart competitive exclusion? Design a garden where two competing species BOTH thrive."*

Watch For

Provide plant data cards with light and nutrient needs. Students design planting arrangements that partition resources.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sunlight Available, Soil Nutrients
- **INTERNAL:** Plant A Growth, Plant B Growth, Resource Depletion

Relationships

- **Sunlight Available to Plant A Growth:** POSITIVE (+) — More available sunlight provides more energy for photosynthesis, allowing Plant A to produce more biomass and grow taller.
- **Sunlight Available to Plant B Growth:** POSITIVE (+) — More available sunlight provides more energy for Plant B's photosynthesis, but this is reduced if Plant A shades Plant B.
- **Soil Nutrients to Plant A Growth:** POSITIVE (+) — More soil nutrients provide the minerals needed for cell growth and reproduction, supporting Plant A's biomass increase.
- **Soil Nutrients to Plant B Growth:** POSITIVE (+) — More soil nutrients support Plant B's growth, but competition with Plant A for the same nutrient pool limits availability.
- **Plant A Growth to Resource Depletion:** POSITIVE (+) — As Plant A grows, it consumes more sunlight (through shading) and more soil nutrients (through root absorption), increasing the rate of resource depletion.
- **Plant B Growth to Resource Depletion:** POSITIVE (+) — As Plant B grows, it also consumes sunlight and nutrients, contributing to overall resource depletion in the shared environment.
- **Resource Depletion to Plant A Growth:** NEGATIVE (-) — As resources are depleted, there is less sunlight and fewer nutrients available for Plant A, slowing its growth rate.
- **Resource Depletion to Plant B Growth:** NEGATIVE (-) — Resource depletion reduces available sunlight and nutrients for Plant B. Because Plant A may already be capturing more, Plant B is disproportionately affected.

Key Discoveries

- Even small initial advantages in resource capture can compound over time, leading to one species dominating
- Competition for light is often asymmetric because taller plants shade shorter ones but not vice versa
- Resource depletion creates a feedback loop where growth reduces available resources, which limits further growth
- Two species can coexist if they partition resources (different root depths, different light needs, different growing seasons)
- The competitive exclusion principle means identical niches cannot support two species long-term

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants cooperate and share resources"	While some mutualistic relationships exist (mycorrhizal networks), plants in close proximity generally compete intensely for light, water, and nutrients. They do not share resources voluntarily. Even trees connected by fungal networks are engaged in complex competitive and cooperative dynamics, not simple sharing.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Competition only matters when resources are scarce"	Competition occurs whenever organisms use the same resource, even if that resource seems abundant. Two plants can compete for sunlight even on a bright day because one shades the other. Competition shapes growth patterns long before resources become critically scarce.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The bigger plant always wins"	While size is a major advantage, other factors matter. A smaller plant with deeper roots might access nutrients the taller plant cannot reach. A shade-tolerant species might thrive in the understory. The 'winner' depends on which resources are most limiting and how each species accesses them.